

***Proof of $2^n - 1$ steps through
Mathematical Induction.***

Say, n is number of disks.

Base Case:

$$T_0 = 2^0 - 1 = 1 - 1 = 0. \text{ [i. e. with 0 disk } n = 0 \text{]}$$

$$T_1 = 2^1 - 1 = 2 - 1 = 1. \text{ [i. e. with 1 disk } n = 1 \text{]}$$

which we can write directly.

Hypothesis:

*Assume for a smaller number of disks k ($k < n$),
the optimal solution requires $2^k - 1$ moves.*

Induction Step:

- *Now consider solving n – disk puzzle.*
 - *Move the top $(k - 1)$ disks to a temporary tower (Auxiliary tower) using their optimal solution $\rightarrow 2^k - 1$ moves.*
 - *Move the largest n^{th} disk to its final destination peg.*

- *Move the $(k - 1)$ disks back to their final destination on top of disk `n` using their reversed optimal solution (another $2^k - 1$ moves.)*

Counting Moves: –

$$\begin{aligned}
 \text{Total moves} &= (2^{k-1} - 1) + 1(2^{k-1} - 1) \\
 &= 2(2^{k-1} - 1) + 1 \\
 &= 2^{k-1+1} - 2 + 1 \quad (a \times a^n = a^{n+1}) \\
 &= 2^k - 1
 \end{aligned}$$

Substituting `k` with `n` gives :

$$\begin{aligned}
 &= 2(2^{n-1} - 1) + 1 \\
 &= 2^{n-1+1} - 2 + 1 \\
 &= 2^n - 1
 \end{aligned}$$

Conclusion: –

Since the hypothesis holds for 1 disk and extends to larger disks through the induction ,we can conclude that the minimal number of moves for the $n - \text{disk}$ Tower of Hanoi is indeed $2^n - 1$.
